

Aden University
Faculty of Engineering
Center of Preparatory Year

Subject: Physics I
Time: 1hr.

Test
Module B1

Day: 3 / 2 /2022
Year: 2021-2022

Name:

Group:

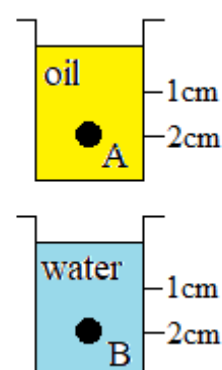
Reg. No.

Part I: For each statement, mark T for true and F for false (12 marks)

1- The Buoyant force is equal to the mass of the displaced fluid	[]	[F]
2- The heat transferred by waves in radiation	[T]	[]
3- The absolute pressure is always more than the gauge a pressure for open fluid	[T]	[]
4- In Pascal application, the output pressure is more than input pressure	[]	[F]
5- The change in Celsius scale is equal to the change in Kelvin scale	[T]	[]
6- The water is the only liquid can causes a buoyant force	[]	[F]

Part II: Select the correct answer from the following sentences (14 marks)

7- The amount of space occupied by the object is			
a) <u>Volume</u>	b) Weight	c) Mass	d) Density
8- When the gas is heated its mass of the gas			
a) Increases	b) Decreases	c) Expanded	d) <u>None of these</u>
9- A thermistor is used in			
a) Calorimeter	b) Fluid thermometer	c) <u>Electronic thermometer</u>	d) None of these
10- A hot metal is placed in water. The heat			
a) <u>Enters into the water</u>	b) Leaves out the water	c) Enters into the metal	d) None of these
11- When the object is partially immersed. The object will			
a) Sink	b) Expand	c) <u>Float</u>	d) None of these
12- Any object immersed in liquid is buoyed up			
a) <u>Archimedes'</u>	b) Zeroth	c) Pascal	d) None of these
13- Two Objects A and B are the same depths. The pressure on object A is			



a) More than B	a) <u>Less than B</u>	a) Same as B	d) None of these
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Part III: Solve the problems to choose the correct answer**(24 marks)**

Gravity of earth	Atmospheric pressure	Water density	Specific heat of water
$g = 10\text{m/s}^2$	$P_{\text{atm}} = 1 \text{ atm} = 10^5 \text{ Pa}$	$\rho = 1000 \text{ kg/m}^3$	$c = 4186 \text{ J/kg.}^\circ\text{C}$

14- A pan of water is heated from 90 °F to 117 °F. The change of the temperature in Fahrenheit scale is:

- | | | | |
|-------|--------------|---------|------------------|
| a) 15 | b) 27 | c) 48.6 | d) None of these |
|-------|--------------|---------|------------------|

15- A copper cube of 20 cm length is heated from 10 °C to 50 °C. The change in area is:

- | | | | |
|--------------------------------|--------------------------|--------------------------|--------------------------|
| a) 3.264 cm² | b) 4.608 cm ² | c) 1.632 cm ² | d) 0.544 cm ² |
|--------------------------------|--------------------------|--------------------------|--------------------------|

16- A 0.05 kg metal is heated to 200 °C and dropped into 0.4 kg water at 20 °C. If the final equilibrium temperature is 22.5 °C, The specific heat of the metal is

- | | | | |
|------------------|------------------|--------------------------|-------------------|
| a) 490.8 J/kg.°C | b) 452.5 J/kg.°C | c) 471.66 J/kg.°C | d) None of theses |
|------------------|------------------|--------------------------|-------------------|

17- A 50 g iron is at 25 °C. If 1200 J is added to it by heat. The final temperature of the iron is:

- | | | | |
|------------|-------------------|----------|------------------|
| a) 51.7 °C | b) 78.6 °C | c) 87 °C | d) None of these |
|------------|-------------------|----------|------------------|

18- 60 N object in air but in water is 45 N. The Buoyant force is:

- | | | | |
|----------|----------------|---------|---------|
| a) 105 N | b) 15 N | c) 45 N | d) 60 N |
|----------|----------------|---------|---------|

19- When 1500 J is added to a metal of heat capacity 30 J/°C its length is increased from 30 m to 30.0255 m. The metal is:

- | | | | |
|------------------|----------|-------------|------------------|
| a) Copper | b) Steel | c) Aluminum | d) None of these |
|------------------|----------|-------------|------------------|

Solution steps are required for question 19

$$\Delta Q = C \Delta T_C \Rightarrow \Delta T_C = \frac{\Delta Q}{C} = \frac{1500}{30} = 50^\circ\text{C}$$

$$\Delta L = \alpha L_i \Delta T_C \Rightarrow \alpha = \frac{\Delta L}{L_i \Delta T_C} = \frac{0.0255}{[30 \times 50]} = 1.7 \times 10^{-5} ^\circ\text{C}^{-1} = 17 \times 10^{-6} ^\circ\text{C}^{-1}$$

20- The gauge pressure in order to suck liquid of density 1800 kg/m³ up a tube by a height of 15 m is:

- | | | | |
|--------------------------------|---------------|---------------------|------------------|
| a) - 2.7 × 10 ⁵ atm | b) - 27000 Pa | c) - 2.7 atm | d) None of these |
|--------------------------------|---------------|---------------------|------------------|

21- A metal of 50 kg in air, 40 kg in water and 425 N in oil. The density of the oil is:

- | | | | |
|--------------------------------|--------------------------|--------------------------|--------------------------|
| a) 750 kg/m³ | b) 820 kg/m ³ | c) 850 kg/m ³ | d) 920 kg/m ³ |
|--------------------------------|--------------------------|--------------------------|--------------------------|

Solution steps are required for question 21

$$m_o = 50 \text{ kg}, m_{o \text{ water}} = 40 \text{ kg}, m_{o \text{ oil}} = 425/10 = 42.5 \text{ kg}$$

- In water: $m_b = m_o - m_{o \text{ water}} = 50 - 40 = 10 \text{ kg}$

$$m_b = \rho_f V_f \Rightarrow V_f = m_b / \rho_f = \frac{10}{1000} = 0.01 \text{ m}^3 \Rightarrow V_o = V_f = V_{\text{oil}} = 0.01 \text{ m}^3$$

- In oil: $m_b = m_o - m_{o \text{ oil}} = 50 - 42.5 = 7.5 \text{ kg}$

$$m_b = \rho_{\text{oil}} V_{\text{oil}} \Rightarrow \rho_{\text{oil}} = m_b / V_{\text{oil}} = \frac{7.5}{0.01} = 750 \text{ kg/m}^3$$

GOOD LUCK

Aden University
Faculty of Engineering
Center of Preparatory Year

Subject: Physics I
Time: 1hr.

Test
Module B3

Day: 1 / 2 /2022
Year: 2021-2022

Name:

Group:

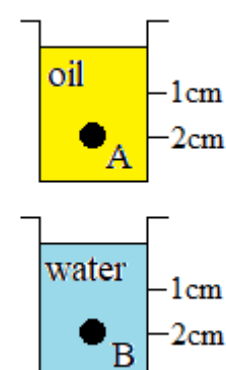
Reg. No.

Part I: For each statement, mark T for true and F for false (12 marks)

1- The Buoyant force is equal to the weight of displaced fluid	[T]	[]
2- The heat is transferred at thermal equilibrium	[]	[F]
3- The absolute pressure is always more than the gauge a pressure for open fluid	[T]	[]
4- The zero Fahrenheit is hotter than zero Kelvin	[T]	[]
5- A density of solid decreases when it is heated	[T]	[]
6- The water is the only liquid can causes a buoyant force	[]	[F]

Part II: Select the correct answer from the following sentences (14 marks)

7- The force exerted by the gravity on the object is			
a) Volume	<u>b) Weight</u>	c) Mass	d) Density
8- When the gas is heated its mass			
a) Increases	b) Decreases	c) Expanded	<u>d) None of theses</u>
9- A thermistor is used in			
a) Calorimeter	b) Fluid thermometer	<u>c) Electronic thermometer</u>	d) None of these
10- The ice is placed in water. The heat			
a) Enters into the water	b) Leaves out the ice	<u>c) Enters into the ice</u>	d) None of these
11- The thermal volume coefficient is			
<u>a) 3α</u>	b) α	c) α^3	d) γ
12- Any change in pressure will transmitted without loss to every part in container			
a) Zeroth	<u>b) Pascal</u>	c) Archimedes'	d) None of these
13- Two Objects A and B are the same depths. The pressure on object B is			



<u>a) More than A</u>	a) Less than A	a) Same as A	d) None of these
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Module B3

Part III: Solve the problems to choose the correct answer

(24 marks)

Gravity of earth $g = 10 \text{ m/s}^2$	Atmospheric pressure $P_{\text{atm}} = 1 \text{ atm} = 10^5 \text{ Pa}$	Water density $\rho = 1000 \text{ kg/m}^3$	Specific heat of water $c = 4186 \text{ J/kg} \cdot ^\circ\text{C}$
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14- A pan of water is heated from 90 °F to 126 °F. The change of the temperature in Fahrenheit scale is:

a) 64.8	b) 20	c) 36	d) None of these
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15- An Aluminum cube of 20 cm length is heated from 10 °C to 50 °C. The change in area is:

a) 2.304 cm ²	b) 4.608 cm²	c) 0.768 cm ²	d) None of these
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16- A 0.05 kg metal is heated to 200 °C and dropped into 0.4 kg water at 20 °C. If the final equilibrium temperature is 22.7 °C, The specific heat of the metal is

a) 452.5 J/kg.°C	b) 509.97 J/kg.°C	c) 529.16 J/kg.°C	d) None of these
------------------	--------------------------	-------------------	------------------

17- A 50 g aluminum is at 25 °C. If 1200 J is added to it by heat. The final temperature of the aluminum is:

a) 51.7 °C	b) 56.1 °C	c) 87 °C	d) None of these
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18- 105 N object in air but in water is 45 N. The Buoyant force is:

a) 105 N	b) 165 N	c) 45 N	d) 60 N
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19- When 1500 J is added to a metal of heat capacity 30 J/°C its length is increased from 30 m to 30.036 m. The metal is:

a) Copper	b) Aluminum	c) Brass	d) None of these
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Solution steps are required for question 19

$$\Delta Q = C \Delta T_c \Rightarrow \Delta T_c = \frac{\Delta Q}{C} = \frac{1500}{30} = 50 ^\circ\text{C}$$

$$\Delta L = \alpha L_i \Delta T_c \Rightarrow \alpha = \frac{\Delta L}{L_i \Delta T_c} = \frac{0.036}{[30 \times 50]} = 2.4 \times 10^{-5} ^\circ\text{C}^{-1} = 24 \times 10^{-6} ^\circ\text{C}^{-1}$$

20- The gauge pressure in order to suck liquid of density 1000 kg/m³ up a tube by a height of 15 m is:

a) - 0.27 atm	b) - 0.15 atm	c) - 1.5 atm	d) None of these
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21- A metal of 50 kg in air, 40 kg in water and 415 N in oil. The density of the oil is:

a) 750 kg/m ³	b) 820 kg/m ³	c) 850 kg/m³	d) 920 kg/m ³
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Solution steps are required for question 21

$$m_o = 50 \text{ kg}, m_{o \text{ water}} = 40 \text{ kg}, m_{o \text{ oil}} = 415/10 = 41.5 \text{ kg}$$

- In water: $m_b = m_o - m_{o \text{ water}} = 50 - 40 = 10 \text{ kg}$

$$m_b = \rho_f V_f \Rightarrow V_f = m_b / \rho_f = \frac{10}{1000} = 0.01 \text{ m}^3 \Rightarrow V_o = V_f = V_{\text{oil}} = 0.01 \text{ m}^3$$

- In oil: $m_b = m_o - m_{o \text{ oil}} = 50 - 41.5 = 8.5 \text{ kg}$

$$m_b = \rho_{\text{oil}} V_{\text{oil}} \Rightarrow \rho_{\text{oil}} = m_b / V_{\text{oil}} = \frac{8.5}{0.01} = 850 \text{ kg/m}^3$$

GOOD LUCK

Aden University
Faculty of Engineering
Center of Preparatory Year

Subject: Physics I
Time: 1.5 hours

Test
Module C4

Day: 6 / 7 /2021
Year: 2020-2021

Name:

Group:

Reg. No.

Part I: For each statement, mark T for true and F for false (16 marks)

- | | |
|---|--------------------|
| 1- In fluid at rest, The pressure is exerted equally in all directions | (<u>T</u>) (F) |
| 2- Celsius scale is based on properties of the water | (<u>T</u>) (F) |
| 3- The weight of the object is constant , it does not change with gravity | (T) (<u>F</u>) |
| 4- An example of radiation is the heat transferred by the waves | (<u>T</u>) (F) |
| 5- The pressure due to a rest fluid is called hydrostatic pressure | (<u>T</u>) (F) |
| 6- A pressure is not affected by a shape of the container | (<u>T</u>) (F) |
| 7- Heating by convection can occur only through gases | (T) (<u>F</u>) |
| 8- A Kelvin scale is based on a correct zero | (<u>T</u>) (F) |

Part II: Select the correct answer from the following sentences (16 marks)

- | | | | | |
|---|----------------------|-------------------------|-------------------------|--------------------------|
| 9- A temperature change in Fahrenheit as compared to a change in Celsius is | (a) Less | (b) <u>More</u> | (c) Same | (d) None of these |
| 10- In a thermometer, the thermistor should be | (a) Tall | (b) Very Large | (c) <u>Very Small</u> | (d) Short |
| 11- A change in resistance due to a temperature change occurred in | (a) Gas thermometer | (b) Mercury thermometer | (c) Alcohol thermometer | (d) <u>None of these</u> |
| 12- When the hot metal is placed in a water, some energy (heat) by water is | (a) Lost | (b) Melt | (c) <u>Gained</u> | (d) None of these |
| 13- If the ratio of the object to the fluid densities is less than one, the object is | (a) <u>Moving up</u> | (b) Sinking | (c) At rest | (d) Floating |
| 14- A measurement unit of the weight is | (a) m | (b) <u>N</u> | (c) kg | (d) m ² |
| 15- A full or partially immersed object in a fluid is buoyed up | (a) Calorimetry | (b) Pascal | (c) Zeroth law | (d) <u>Archimedes'</u> |
| 16- In Pascal application, the pressures should be | (a) Closed | (b) Rest | (c) <u>Same</u> | (d) Open |

Part III: Solve the problems to choose the correct answer (18 marks)

Take $g = 10 \text{ m/s}^2$ and $P_{\text{atm}} = 1 \text{ atm} = 10^5 \text{ Pa}$

- | | | | | |
|---|------------------|-----------|---------------|-------------------|
| 17- The change of temperature is 30 K. The equivalent change in Fahrenheit scale is | (a) 30 | (b) 16.7 | (c) <u>54</u> | (d) None of these |
| 18- A 50 g sample of copper is at 30 °C. If 1200 J of energy is added to it by heat, The final temperature of the copper is ($c = 387 \text{ J/kg.}^\circ\text{C}$) | (a) <u>92 °C</u> | (b) 62 °C | (c) 32 °C | (d) None of these |

Module C4

-
- 19- Two metals with different length are L_1 and L_2 have $\alpha_1 = 10 \times 10^{-6} \text{ } ^\circ\text{C}^{-1}$ and $\alpha_2 = 14 \times 10^{-6} \text{ } ^\circ\text{C}^{-1}$.
If $L_1 - L_2 = 12 \text{ cm}$ at any temperatures. Then L_1 is equal to
(a) 48 cm (b) 36 cm (c) 27 cm **(d) 42 cm**
-
- 20- A 5 kg object is released from rest while fully immersed in a liquid. The liquid displaced has a mass of 2 kg.
The acceleration of the object is
(a) 6 m/s² (b) 4 m/s² (c) 4.6 m/s² (d) None of these
-
- 21- A 50 N object floats in liquid $\rho_f = 10^3 \text{ kg/m}^3$. If the object is immersed by 4 cm and object's height above a surface of liquid is 1 cm. The volume of the object should be
(a) $8.3 \times 10^{-3} \text{ m}^3$ **(b) $6.25 \times 10^{-3} \text{ m}^3$** (c) $7.5 \times 10^{-3} \text{ m}^3$ (d) None of these
-
- 22- A piece of metal of $m_o = 0.5 \text{ kg}$ and a density $\rho_o = 8.7 \times 10^3 \text{ kg/m}^3$. The metal is immersed in fluid of density $\rho_f = 10^3 \text{ kg/m}^3$. The weight of the metal in fluid is
(a) 4.36 N (b) 5 N **(c) 4.4 N** (d) None of these
-

Good Luck

Aden University
Faculty of Engineering
Center of Preparatory Year

Subject: Physics I
Time: 1hr.

Test
Module B4

Day: 3 / 2 /2022
Year: 2021-2022

Name:

Group:

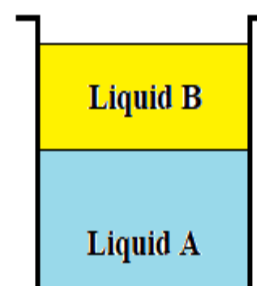
Reg. No.

Part I: For each statement, mark T for true and F for false (12 marks)

1- The weight of the displaced fluid is called a Buoyant force	☐ T	☐
2- The pressure is same in Pascal applications	☐ T	☐
3- The absolute pressure is always more than the gauge a pressure for open fluid	☐ T	☐
4- The object is sinking when it is partially immersed	☐	☐ F
5- A density of solid increases when it is heated	☐	☐ F
6- The water is the only liquid can causes a buoyant force	☐	☐ F

Part II: Select the correct answer from the following sentences (14 marks)

7- The mass per unit volume is			
a) Volume	b) Weight	c) Mass	<u>d) Density</u>
8- When the gas is heated its mass			
a) Increases	b) Decreases	c) Expanded	<u>d) None of these</u>
9- A digital thermometer uses			
a) Liquid	<u>b) Thermistor</u>	c) Gas	d) None of these
10- The ice is placed in water. The heat			
<u>a) Leaves out the water</u>	b) Leaves out the ice	c) Enters into the water	d) None of these
11- The surface area coefficient is			
a) β	b) α	c) α^2	<u>d) 2α</u>
12- The amount of heat needed to increase a temperature of object by 1°C is			
<u>a) Heat capacity</u>	b) Specific heat	c) Calorimetry	d) None of these
13- Two liquids in container. The liquid B has			



<u>a) Less density than A</u>	b) More density than A	c) Same density as A	d) None of these
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Part III: Solve the problems to choose the correct answer**(24 marks)**

Gravity of earth	Atmospheric pressure	Water density	Specific heat of water
$g = 10 \text{ m/s}^2$	$P_{\text{atm}} = 1 \text{ atm} = 10^5 \text{ Pa}$	$\rho = 1000 \text{ kg/m}^3$	$c = 4186 \text{ J/kg}^\circ\text{C}$

14- A pan of water is heated from 90 °F to 135 °F. The change of the temperature in Fahrenheit scale is:

- a) 81 **b) 45** c) 25 d) None of these

15- A brass cube of 20 cm length is heated from 10 °C to 50 °C. The change in area is:

- a) 1.824 cm² b) 0.608 cm² **c) 3.648 cm²** d) None of these

16- 0.05 kg metal is heated to 200 °C and dropped into 0.4 kg water at 20 °C. If the final equilibrium temperature is 22.8 °C, The specific heat of the metal is

- a) 452.5 J/kg.°C b) 509.97 J/kg.°C **c) 529.16 J/kg.°C** d) None of these

17- A 50 g iron is at 2.5 °C. If 1200 J is added to it by heat. The final temperature of the iron is:

- a) 51.7 °C **b) 56.1 °C** c) 87 °C d) None of these

18- 35 N object in air but in water is 20 N. The Buoyant force is:

- a) 20 N b) 75 N **c) 15 N** d) 35 N

19- When 1500 J is added to a metal of heat capacity 30 J/°C its length is increased from 30 m to 30.0285 m. The metal is:

- a) Brass** b) Steel c) Aluminum d) None of these

Solution steps are required for question 19

$$\Delta Q = C \Delta T_C \Rightarrow \Delta T_C = \frac{\Delta Q}{C} = \frac{1500}{30} = 50^\circ\text{C}$$

$$\Delta L = \alpha L_i \Delta T_C \Rightarrow \alpha = \frac{\Delta L}{L_i \Delta T_C} = \frac{0.0285}{[30 \times 50]} = 1.9 \times 10^{-5} \text{ }^\circ\text{C}^{-1} = 19 \times 10^{-6} \text{ }^\circ\text{C}^{-1}$$

20- The gauge pressure in order to suck liquid of density 1000 kg/m³ up a tube by a height of 1.5 m is:

- a) - 0.27 atm **b) - 0.15 atm** c) - 15000 atm d) None of these

21- A metal of 50 kg in air, 40 kg in water and 41.8 kg in oil. The density of the oil is:

- a) 750 kg/m³ **b) 820 kg/m³** c) 850 kg/m³ d) 920 kg/m³

Solution steps are required for question 21

$$m_o = 50 \text{ kg}, m_{o \text{ water}} = 40 \text{ kg}, m_{o \text{ oil}} = 41.8 \text{ kg}$$

- In water: $m_b = m_o - m_{o \text{ water}} = 50 - 40 = 10 \text{ kg}$

$$m_b = \rho_f V_f \Rightarrow V_f = m_b / \rho_f = \frac{10}{1000} = 0.01 \text{ m}^3 \Rightarrow V_o = V_f = V_{\text{oil}} = 0.01 \text{ m}^3$$

- In oil: $m_b = m_o - m_{o \text{ oil}} = 50 - 41.8 = 8.2 \text{ kg}$

$$m_b = \rho_{\text{oil}} V_{\text{oil}} \Rightarrow \rho_{\text{oil}} = m_b / V_{\text{oil}} = \frac{8.2}{0.01} = 820 \text{ kg/m}^3$$

GOOD LUCK

Aden University
Faculty of Engineering
Center of Preparatory Year

Subject: Physics I
Time: 1hr. and 15 min.

Test
Module A3

Day: 31 / 1 /2022
Year: 2021-2022

Name:

Group:

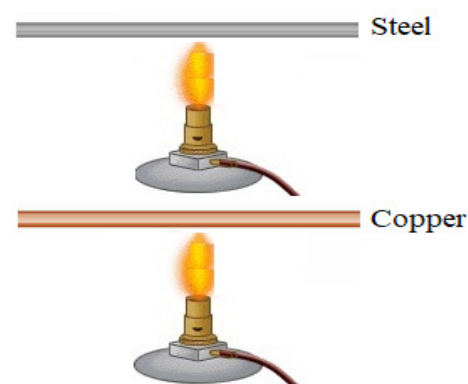
Reg. No.

Part I: For each statement, mark T for true and F for false (15 marks)

1- A distance between molecules decreases when a solid changes to liquid	[]	[F]
2- Liquids and gases can take a shape of container	[T]	[]
3- A calorimeter is used to measure a heat transfer	[T]	[]
4- We can make a plastic sink and a steel float in water	[T]	[]
5- The mass of object decreases at the moon	[]	[F]
6- A balloon moving up in the air is based on Archimedes' principle	[T]	[]

Part II: Select the correct answer from the following sentences (14 marks)

7- The absolute zero temperature is the same as			
a) Zero Celsius	b) Zero Fahrenheit	<u>c) Zero Kelvin</u>	d) None of these
8- Which of the following heat transfer types has no thermal contact			
a) Conduction	b) Radiation	c) convection	<u>d) None of these</u>
9- The form of energy that causes a hotness or coldness is called			
a) Pressure	b) Kinetic Energy	c) Temperature	<u>d) Heat</u>
10- The pressure is the same in every part of the fluid in a closed container, this according to			
<u>a) Pascal</u>	b) Archimedes	c) Zeroth	d) None of these
11- Any thermometer has			
a) Infinite range	<u>b) limited range</u>	c) no limited range	d) None of these
12- The pressure due to fluid is directed			
a) Downward	b) Upward	<u>c) In all directions</u>	d) None of these
13- Steel and Copper rods shown in the figure are heated to the same temperature. The increase of the copper is:			



<u>a) More than steel</u>	b) Less than steel	c) Same as steel	d) None of these
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Module A3

Part III: Solve the problems to choose the correct answer

(24 marks)

Gravity of earth $g = 10 \text{ m/s}^2$	Atmospheric pressure $P_{\text{atm}} = 1 \text{ atm} = 10^5 \text{ Pa}$	Water density $\rho = 1000 \text{ kg/m}^3$	Specific heat of water $c = 4186 \text{ J/kg} \cdot ^\circ\text{C}$
14- A water is heated from 90 °F to 126 °F. The change in the Kelvin scale is:			
a) 64.8	b) 20	c) 36	d) 293
15- A brass of 60 cm is heated from 10 °C to 50 °C. The final length of the brass is:			
a) 60.0456 cm	b) 0.0576 cm	c) 60.0576 cm	d) 0.0456
16- A 1.5 kg iron initially at 600 °C is dropped into a bucket containing 20.0 kg of water at 40 °C. The final temperature is:			
a) 34.12°C	b) 29.6 °C	c) 44.46 °C	d) None of these
17- The heat required to increase a temperature of 30 g of iron by 40 °C is:			
a) 537.6 J	b) 537.6 kJ	c) 537600 J	d) None of these
18- 50 kg object displaces 30 kg of water. The Buoyant force is			
a) 200 N	b) 20 N	c) 30 N	d) 300 N
19- Two objects, object A at 550 °C and object B at 400 °C are in thermal contact. The heat capacity of the object A is twice as the object B. The final thermal equilibrium temperature is			
a) 75 °C	a) 500 °C	a) 450 °C	a) 475 °C

Solution steps are required for question 19

$$Q_B = -\Delta Q_A \Rightarrow C_B \Delta T_B = -C_A \Delta T_A$$

$$C_B (T_f - T_B) = -C_A (T_f - T_A) \Rightarrow C_B (T_f - T_B) = C_A (T_A - T_f)$$

$$\because C_A = 2 C_B \Rightarrow \therefore C_B (T_f - 400) = 2C_B (550 - T_f) \Rightarrow T_f - 400 = 1100 - 2T_f$$

$$3T_f = 1500 \Rightarrow T_f = 1500/3 = 500^\circ\text{C}$$

20- The gauge pressure at depth of 20 m in water (open to atmosphere) is:			
a) 3 pa	b) 3 atm	c) $2 \times 10^5 \text{ atm}$	d) 2 atm
21- 7 kg is added to an object of density $\rho_o = 600 \text{ kg/m}^3$ to be fully immersed in water. The mass of the object is:			
a) 9 kg	b) 10.5 kg	c) 6 kg	d) None of these

Solution steps are required for question 21

$$\text{Object is less density but after added 7 kg: } 7 = m_f - m_o \Rightarrow 7 = m_o \frac{\rho_f}{\rho_o} - m_o$$

$$7 = m_o \left(\frac{\rho_f}{\rho_o} - 1 \right) = m_o \left(1 - \frac{1000}{600} \right) = m_o (0.667) \Rightarrow m_o = 7/0.667 = 10.5 \text{ kg}$$

GOOD LUCK

Aden University Faculty of Engineering Center of Preparatory Year		
Subject: Physics I Time: 1.5 hours	Test Module B2	Day: 5 / 7 /2021 Year: 2020-2021

Name:	Group:	Reg. No.
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Part I: For each statement, mark T for true and F for false (16 marks)

- | | |
|---|--------------------|
| 1- A ratio of the heat capacity to the specific heat is always constant | (<u>T</u>) (F) |
| 2- The heat transfer occurs in solid is by convection | (T) (<u>F</u>) |
| 3- The heat is positive when it is transmitted into the object | (<u>T</u>) (F) |
| 4- The kelvin temperature scale has negative values | (T) (<u>F</u>) |
| 5- The object is sinking when it is moving up in the fluid | (T) (<u>F</u>) |
| 6- A density is the mass per unit volume that has Si unit kg/m ³ | (<u>T</u>) (F) |
| 7- More force is required to lift an object in air than in water | (<u>T</u>) (F) |
| 8- Water floats on oil because, the water is less density than the oil | (T) (<u>F</u>) |

Part II: Select the correct answer from the following sentences (16 marks)

- | | | | | |
|---|-------------------------------------|-------------------------------------|--|-------------------------------------|
| 9- The Zero Celsius is equal to | (a) – 273 K | (b) - 273 °F | (c) - 273 °C | (d) <u>None of these</u> |
| 10- The measurement unit of Specific Heat Capacity is | (a) <u>J/kg.°C</u> | (b) J/°C | (c) °C ⁻¹ | (d) °C |
| 11- The Heat can be exchanged when | (a) T ₁ = T ₂ | (b) T ₁ > T ₂ | (c) <u>T₁ ≠ T₂</u> | (d) T ₁ < T ₂ |
| 12- A Temperature is increased by 1 °C. This will equal to | (a) 274 K | (b) 33.8 °F | (c) 18 °F | (d) <u>1 K</u> |
| 13- The Volume expansion coefficient is | (a) <u>β</u> | (b) 3β | (c) 3β/2 | (d) None of these |
| 14- When the altitude decreases, the pressure will be | (a) Decreased | (b) <u>Increased</u> | (c) Unchanged | (d) None of these |
| 15- The amount of matter in an object is called | (a) Weight | (b) Density | (c) Volume | (d) <u>Mass</u> |
| 16- According to Pascal principle, the area of the output piston as compared to the input piston is | (a) Smaller | (b) <u>Bigger</u> | (c) Same | (d) None of these |

Part III: Solve the problems to choose the correct answer (18 marks)

Take $g = 10 \text{ m/s}^2$ and $P_{\text{atm}} = 1 \text{ atm} = 10^5 \text{ Pa}$

- | | | | | |
|--|---|-----------------------|-----------------|-------------------|
| 17- On a day when the temperature reaches 14 °F. The temperature in degrees Celsius is | (a) 25.6 | (b) - 15 | (c) <u>- 10</u> | (d) - 32.4 |
| 18- A steel rod of 40 cm long at 40 °C is heated to 100°C. The final length is | $\alpha = 11 \times 10^{-6} \text{ } ^\circ\text{C}^{-1}$ (for steel) | | | |
| | (a) 39.9736 cm | (b) <u>40.0264 cm</u> | (c) 0.0264 cm | (d) None of these |

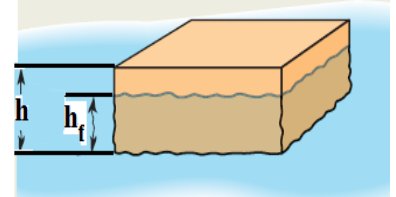
Module B2

19- A certain metal rod has a length of 10 m at 100 °C and a length of 10.04 m at 500 °C. Its length at 200 °C is
(a) 10.01 m (b) 9.99 m (c) 10.02 m (d) 9.98 m

20- If a liquid of density 1000 kg/m³. The gauge pressure at h = 30 m is
(a) 30000 Pa (b) 3 atm (c) 4 atm (d) 40000 Pa

21- A metal has a weight of 5 N is fully immersed in a fluid of density $\rho_f = 960 \text{ kg/m}^3$, then a metal weight is reduced by 2 N. The volume of the metal is
(a) 480 m³ (b) $5.208 \times 10^{-4} \text{ m}^3$ (c) $2.1 \times 10^{-4} \text{ m}^3$ (d) None of these

22- In a Fig. a block of density $\rho_o = 900 \text{ kg/m}^3$ floats face down in a fluid of density $\rho_f = 1200 \text{ kg/m}^3$. The block has 6 cm height. The height of the block immersed is



(a) 4 cm (b) 4.5 cm (c) 4.8 cm (d) None of these

Good Luck

Aden University Faculty of Engineering Center of Preparatory Year		
Subject: Physics I Time: 1.5 hours	Test Module B1	Day: 5 / 7 /2021 Year: 2020-2021

Name:	Group:	Reg. No.
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Part I: For each statement, mark T for true and F for false (16 marks)

- | | |
|---|---------|
| 1- A ratio of the heat capacity to the specific heat is always constant | (T) (F) |
| 2- The heat transfer occurs in the water is by convection | (T) (F) |
| 3- The heat is positive when it is transmitted out of the object | (T) (F) |
| 4- The kelvin temperature scale has not negative values | (T) (F) |
| 5- The object is sinking when it is moving down in fluid | (T) (F) |
| 6- Temperature is one form of energy that has Si unit Kelvin | (T) (F) |
| 7- More force is required to lift an object in water than in air | (T) (F) |
| 8- Oil floats on water because, the oil is less density than the water | (T) (F) |

Part II: Select the correct answer from the following sentences (16 marks)

- | | | | | |
|---|--|-------------------------------------|-------------------------------------|-------------------------------------|
| 9- The Zero Celsius is equal to | (a) – 273 K | (b) - 273 °F | (c) - 273 °C | (d) <u>None of these</u> |
| 10- The measurement unit of Heat Capacity is | (a) J/kg.°C | (b) <u>J/°C</u> | (c) °C ⁻¹ | (d) °C |
| 11- Thermal equilibrium is occurred when | (a) <u>T₁ = T₂</u> | (b) T ₁ ≠ T ₂ | (c) T ₁ > T ₂ | (d) T ₁ < T ₂ |
| 12- A Temperature is 1 °C. This will equal to | (a) <u>274 K</u> | OR | (b) <u>33.8 °F</u> | (c) 1.8 K |
| | | | | (d) 1 K |
| 13- The surface area expansion coefficient is | (a) 2γ | (b) <u>γ</u> | (c) 2γ/3 | (d) None of these |
| 14- When the altitude increases, the pressure will be | (a) <u>Decreased</u> | (b) Increased | (c) Unchanged | (d) None of these |
| 15- The amount of space an object takes is called | (a) Weight | (b) Density | (c) <u>Volume</u> | (d) Pressure |
| 16- According to Pascal principle, the area of the input piston as compared to the output piston is | (a) <u>Smaller</u> | (b) Bigger | (c) Same | (d) None of these |

Part III: Solve the problems to choose the correct answer (18 marks)

Take $g = 10 \text{ m/s}^2$ and $P_{\text{atm}} = 1 \text{ atm} = 10^5 \text{ Pa}$

- | | | | | |
|---|---|-----------------|-----------------------|-------------------|
| 17- On a day when the temperature reaches 5 °F. The temperature in degrees Celsius is | (a) 41 | (b) <u>- 15</u> | (c) - 48.6 | (d) None of these |
| 18- A steel rod of 40 cm long at 40 °C is heated to 80°C. The final length is | | | | |
| | $\alpha = 11 \times 10^{-6} \text{ } ^\circ\text{C}^{-1}$ (for steel) | | | |
| | (a) 39.9824 cm | (b) 0.0176 cm | (c) <u>40.0176 cm</u> | (d) None of these |

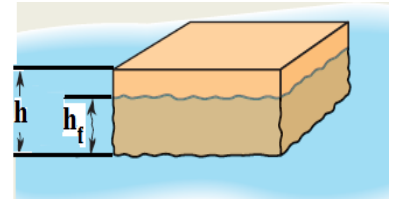
Module B1

19- A certain metal rod has a length of 10 m at 100 °C and a length of 10.04 m at 500 °C. Its length at 0 °C is
(a) 10.01 m **(b) 9.99 m** (c) 10.02 m (d) 9.98 m

20- If a liquid of density 1000 kg/m³. The gauge pressure at h = 10 m is
(a) 10000 Pa (b) 2 atm **(c) 1 atm** (d) 20000 Pa

21- A metal has a weight of 5 N is fully immersed in a fluid of density $\rho_f = 900 \text{ kg/m}^3$, then a metal weight is reduced by 2 N. The volume of the metal is
(a) $2.2 \times 10^{-4} \text{ m}^3$ (b) $3.3 \times 10^{-4} \text{ m}^3$ (c) 450 m³ (d) None of these

22- In a Fig. a block of density $\rho_o = 800 \text{ kg/m}^3$ floats face down in a fluid of density $\rho_f = 1000 \text{ kg/m}^3$. The block has 6 cm height. The height of the block immersed is



(a) 6 cm (b) 4 cm **(c) 4.8 cm** (d) None of these

Good Luck

Aden University Faculty of Engineering Center of Preparatory Year		
Subject: Physics I Time: 1.5 hours	Test Module B3	Day: 6 / 7 /2021 Year: 2020-2021

Name:	Group:	Reg. No.
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Part I: For each statement, mark T for true and F for false (16 marks)

1- A ratio of the heat capacity to the specific heat is always constant	(<u>T</u>) (F)
2- Heat transferred from a cold object into a hot object	(T) (<u>F</u>)
3- The heat is zero when the two objects are not in rest	(T) (<u>F</u>)
4- The zero kelvin can not be reached	(<u>T</u>) (F)
5- The object is partially immersed when its volume is not equal to the volume of fluid	(<u>T</u>) (F)
6- A force per unit volume is the pressure that has Si unit Pa	(T) (<u>F</u>)
7- A pressure due to fluid in rest is transmitted equally in all directions	(<u>T</u>) (F)
8- The height of the fluid is directly proportional to the density	(T) (<u>F</u>)

Part II: Select the correct answer from the following sentences (16 marks)

9- The Zero Celsius is equal to	(a) – 273 K	(b) - 273 °F	(c) - 273 °C	(d) <u>None of these</u>
10- The measurement unit of Celsius scale is	(a) J/kg.°C	(b) J/°C	(c) 1/°C	(d) <u>°C</u>
11- When the heat is transferred from surrounding into an object, the heat is	(a) Negative	(b) <u>Positive</u>	(c) Zero	(d) None of these
12- A Temperature is 1 °C. This will equal to	(a) <u>274 K</u>	Or (b) <u>33.8 °F</u>	(c) 1.8 °F	(d) 1 K
13- Energy required by a substance to raise temperature by 1 °C is called	(a) 1.8 °F	(b) Specific heat	(c) <u>Heat capacity</u>	(d) None of these
14- The boiling point of water is	(a) 0 °C	(b) <u>373 K</u>	(c) 122 °F	(d) None of these
15- According to Pascal principle, the container should be	(a) <u>Closed</u>	(b) Rest	(c) Open	(d) None of these
16- When a temperature raised, the density is	(a) Unchanged	(b) Increased	(c) <u>Decreased</u>	(d) None of these

Part III: Solve the problems to choose the correct answer (18 marks)

Take $g = 10 \text{ m/s}^2$ and $P_{\text{atm}} = 1 \text{ atm} = 10^5 \text{ Pa}$

17- On a day when the temperature reaches 23 °F. The temperature in degrees Celsius is	(a) 99	(b) - 16.2	(c) 30.6	(d) <u>- 5</u>
18- A steel rod of 40 cm long at 40 °C is heated to 90°C. The final length is $\alpha = 11 \times 10^{-6} \text{ } ^\circ\text{C}^{-1}$ (for steel)	(a) 0.022 cm	(b) 39.9 cm	(c) <u>40.022 cm</u>	(d) None of these

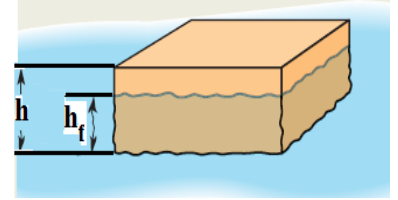
Module B3

19- A certain metal rod has a length of 10 m at 100 °C and a length of 10.04 m at 500 °C. Its length at -100 °C is
(a) 10.01 m (b) 9.99 m (c) 10.02 m **(d) 9.98 m**

20- If a liquid of density 1000 kg/m³. The gauge pressure at h = 40 m is
(a) 4 atm (b) 5 atm (c) 40000 (d) 50000 Pa

21- A metal has a weight of 5 N is fully immersed in a fluid of density $\rho_f = 840 \text{ kg/m}^3$, then a metal weight is decreased by 2 N. The volume of the metal is
(a) 420 m³ **(b) $2.4 \times 10^{-4} \text{ m}^3$** (c) $3.571 \times 10^{-4} \text{ m}^3$ (d) None of these

22- In a Fig. a block of density $\rho_o = 900 \text{ kg/m}^3$ floats face down in a fluid of density $\rho_f = 1000 \text{ kg/m}^3$. The block has 6 cm height. The height of the block immersed is



(a) 5.4 cm (b) 4 cm (c) 4.8 cm (d) None of these

Good Luck

Aden University Faculty of Engineering Center of Preparatory Year		
Subject: Physics I Time: 1.5 hours	Test Module A2	Day: 5 / 7 /2021 Year: 2020-2021

Name:	Group:	Reg. No.
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Part I: For each statement, mark T for true and F for false (16 marks)

- | | |
|--|--------------------|
| 1- The pressure is zero as a temperature reached 32 °F | (T) (<u>F</u>) |
| 2- If two objects are in thermal equilibrium, they must have the same temperatures | (<u>T</u>) (F) |
| 3- Generally, liquids expand less than solids for the same temperature change | (T) (<u>F</u>) |
| 4- The zero point of Celsius scale is below the zero point of Fahrenheit scale | (T) (<u>F</u>) |
| 5- 100 kPa. is approximately equal to 1 atm. | (<u>T</u>) (F) |
| 6- The object floats in a fluid is accelerated downward | (T) (<u>F</u>) |
| 7- A shape of the object will determine if the object sink or float | (T) (<u>F</u>) |
| 8- An object with the less density than a fluid will float | (<u>T</u>) (F) |

Part II: Select the correct answer from the following sentences (16 marks)

- | | | | | |
|---|----------------------------|------------------------------|------------------------|-------------------------------|
| 9- the Si unit of a pressure is | (a) Kelvin | (b) Celsius | (c) Fahrenheit | (d) <u>Pascal</u> |
| 10- The ratio of the input to the output forces in Pascal law is | (a) Equal to 1 | (b) More than 1 | (c) <u>Less than 1</u> | (d) None of these |
| 11- A measure of an average kinetic energy of particles is called | (a) <u>Temperature</u> | (b) Heat | (c) Pressure | (d) None of these |
| 12- A transfer of heat when no thermal contact is needed is called | (a) Conduction | (b) Radiation | (c) Convection | (d) <u>None of these</u> |
| 13- A conversion from Fahrenheit to Celsius is | (a) $\frac{9}{5} T_F + 32$ | (b) $\frac{9}{5} (T_F - 32)$ | (c) $T_F - 32/1.8$ | (d) <u>None of these</u> |
| 14- A density of the object will increase if the volume of the object | (a) <u>Decreases</u> | (b) Increase | (c) Unchanged | (d) None of these |
| 15- In measurement of pressure, the closed end tube is | (a) Thermometer | (b) Manometer | (c) <u>Barometer</u> | (d) Calorimeter |
| 16- The weight of the displaced liquid by object is given by | (a) $\rho_f V_f h_f$ | (b) $m_o g$ | (c) $\rho_o V_o h_o$ | (d) <u>$m_f g$</u> |

Part III: Solve the problems to choose the correct answer (18 marks)

Take $g = 10 \text{ m/s}^2$ and $P_{\text{atm}} = 1 \text{ atm} = 10^5 \text{ Pa}$

- | | | | | |
|--|---|----------------|----------------------|-------------------|
| 17- On a day when the temperature reaches 23 °F. The temperature in Kelvin scale is | (a) -5 | (b) -16.2 | (c) <u>268</u> | (d) None of these |
| 18- A steel rod of 40 cm long at 40 °C is heated to 100°C. The length should be increased by | $\alpha = 11 \times 10^{-6} \text{ } ^\circ\text{C}^{-1}$ (for steel) | | | |
| | (a) 39.9736 cm | (b) 40.0264 cm | (c) <u>0.0264 cm</u> | (d) None of these |

Module A2

19- Metal and Steel are at the same lengths at 20 °C, when the they are heated the same temperature, the metal increased by 0.048 cm and the Steel increased by 0.022 cm. The metal is

- (a) Copper **(b) Aluminum** (c) Brass (d) None of these

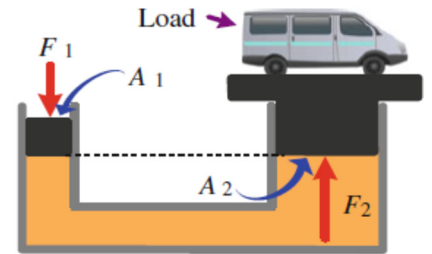
20- To suck liquid of density 1000 kg/m³ up a straw to a maximum height of 4.0 m. The minimum gauge pressure must you produce is

- (a) – 392 Pa (b) 392 Pa (c) 40000 Pa **(d) - 40000 Pa**

21- A metal has a weight of 5 N is fully immersed in a fluid of density $\rho_f = 960 \text{ kg/m}^3$, then a metal weight is decreased to 2 N. The volume of the metal is

- (a) $3.125 \times 10^{-4} \text{ m}^3$** (b) $5.208 \times 10^{-4} \text{ m}^3$ (c) 480 m^3 (d) None of these

22- For the car lift shown. The pistons on the left and right have areas 25 cm² and 75 cm² respectively. The car has a total weight of 15000 N. The force must be applied on the input piston is



- (a) 50 N (b) 500 N (c) 450000 N **(d) 5000 N**

Good Luck

Aden University Faculty of Engineering Center of Preparatory Year		
Subject: Physics I Time: 1.5 hours	Test Module A1	Day: 5 / 7 /2021 Year: 2020-2021

Name:	Group:	Reg. No.
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Part I: For each statement, mark T for true and F for false (16 marks)

- | | |
|--|--------------------|
| 1- The pressure is zero as a temperature reached 0 °C | (T) (<u>F</u>) |
| 2- If two objects are in thermal contact, they must have the same temperatures | (T) (<u>F</u>) |
| 3- Generally, liquids expand more than solids for the same temperature change | (<u>T</u>) (F) |
| 4- The zero point of Celsius scale is above the zero point of Fahrenheit scale | (<u>T</u>) (F) |
| 5- 100 Pa is approximately equal to 1 atm. | (T) (<u>F</u>) |
| 6- The buoyant force of fluid is directed downward | (T) (<u>F</u>) |
| 7- A shape of the object will not determine if the object sink or float | (<u>T</u>) (F) |
| 8- An object with the same density as a fluid will float | (T) (<u>F</u>) |

Part II: Select the correct answer from the following sentences (16 marks)

- | | | | | |
|---|----------------------|-------------------------------|--|--------------------------|
| 9- the Si unit of a temperature is | (a) <u>Kelvin</u> | (b) Celsius | (c) Fahrenheit | (d) Pascal |
| 10- The ratio of the output to the input forces in Pascal law is | (a) Equal to 1 | (b) <u>More than 1</u> | (c) Less than 1 | (d) None of these |
| 11- A measure of a total kinetic energy of particles is called | (a) Temperature | (b) Heat | (c) Pressure | (d) <u>None of these</u> |
| 12- A transfer of heat in liquids and gases is according to | (a) Conduction | (b) Radiation | (c) <u>Convection</u> | (d) None of these |
| 13- A conversion from Celsius to Fahrenheit is | (a) $1.8 (T_C + 32)$ | (b) $\frac{5}{9} T_C + 32$ | (c) <u>$\frac{9}{5} T_C + 32$</u> | (d) None of these |
| 14- A density of the object will decrease if the volume of the object | (a) Decreases | (b) <u>Increase</u> | (c) Unchanged | (d) None of these |
| 15- In measurement of pressure, the open end tube is | (a) Thermometer | (b) Calorimeter | (c) Barometer | (d) <u>Manometer</u> |
| 16- The weight of the object placed in liquid is given by | (a) $\rho_f V_f$ | (b) <u>$m_o g$</u> | (c) $\rho_o V_o$ | (d) $m_f g$ |

Part III: Solve the problems to choose the correct answer (18 marks)

Take $g = 10 \text{ m/s}^2$ and $P_{\text{atm}} = 1 \text{ atm} = 10^5 \text{ Pa}$

- | | | | | |
|---|--|----------------------|----------------|-------------------|
| 17- On a day when the temperature reaches 14 °F. The temperature in Kelvin scale is | (a) <u>263</u> | (b) -10 | (c) -32.4 | (d) None of these |
| 18- A steel rod of 40 cm long at 40 °C is heated to 80°C. The length should be increased by | $\alpha = 11 \times 10^{-6} \text{ }^\circ\text{C}^{-1}$ (for steel) | | | |
| | (a) 39.9824 cm | (b) <u>0.0176 cm</u> | (c) 40.0176 cm | (d) None of these |

Module A1

19- Metal and Steel are at the same lengths at 20 °C, when the they are heated to the same temperature, the metal increased by 0.038 cm and the Steel increased by 0.022 cm. The metal is

- (a) Copper (b) Aluminum (c) Brass (d) None of these

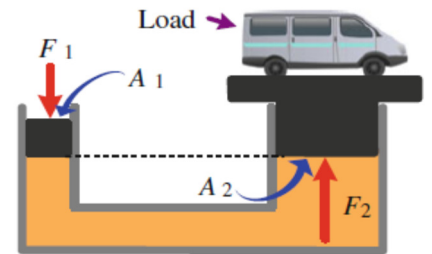
20- To suck liquid of density 1000 kg/m³ up a straw to a maximum height of 0.4 m. The minimum gauge pressure must you produce is

- (a) – 392 Pa (b) 392 Pa (c) - 4000 Pa (d) - 40000 Pa

21- A metal has a weight of 5 N is fully immersed in a fluid of density $\rho_f = 900 \text{ kg/m}^3$, then a metal weight is decreased to 2 N. The volume of the metal is

- (a) $5.6 \times 10^{-4} \text{ m}^3$ (b) $3.3 \times 10^{-4} \text{ m}^3$ (c) 450 m³ (d) None of these

22- For the car lift shown. The pistons on the left and right have areas 2.5 cm² and 750 cm² respectively. The car has a total weight of 15000 N. The force must be applied on the input piston is



- (a) 50 N (b) 500 N (c) 450000 N (d) 5000 N

Good Luck

Aden University Faculty of Engineering Center of Preparatory Year		
Subject: Physics I Time: 1.5 hours	Test Module B4	Day: 5 / 7 /2021 Year: 2020-2021

Name:	Group:	Reg. No.
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Part I: For each statement, mark T for true and F for false (16 marks)

- | | |
|---|--------------------|
| 1- A ratio of the heat capacity to the specific heat is always constant | (<u>T</u>) (F) |
| 2- Heat transferred from a hot object into a cold object | (<u>T</u>) (F) |
| 3- The heat is zero when the two objects are in rest | (T) (<u>F</u>) |
| 4- The zero kelvin can be reached not rapidly | (T) (<u>F</u>) |
| 5- The object is fully immersed when its volume is equal to the volume of fluid | (<u>T</u>) (F) |
| 6- A force per unit area is the pressure that has Si unit Pa | (<u>T</u>) (F) |
| 7- A pressure due to applied force is transmitted equally in all directions | (T) (<u>F</u>) |
| 8- The height of the fluid is inversely proportional to the density | (<u>T</u>) (F) |

Part II: Select the correct answer from the following sentences (16 marks)

- | | | | | |
|---|---------------------|--------------------------|------------------------|--------------------------|
| 9- The Zero Celsius is equal to | (a) – 273 K | (b) - 273 °F | (c) - 273 °C | (d) <u>None of these</u> |
| 10- The measurement unit of expansion coefficient is | (a) J/kg.°C | (b) J/°C | (c) <u>1/°C</u> | (d) °C |
| 11- When the heat is transferred from an object out to a surrounding, the heat is | (a) <u>Negative</u> | (b) Positive | (c) Zero | (d) None of these |
| 12- Average kinetic energy of particles is | (a) Pressure | (b) Heat | (c) <u>Temperature</u> | (d) None of these |
| 13- Energy required per kg of substance to raise temperature by 1 °C is called | (a) 1.8 °F | (b) <u>Specific heat</u> | (c) Heat capacity | (d) None of these |
| 14- The boiling point of water is | (a) 0 °C | (b) 273 K | (c) <u>212 °F</u> | (d) None of these |
| 15- According to Pascal principle, the fluid should be | (a) Closed | (b) <u>Rest</u> | (c) Open | (d) None of these |
| 16- When a temperature raised, the volume is | (a) Unchanged | (b) <u>Increased</u> | (c) Decreased | (d) None of these |

Part III: Solve the problems to choose the correct answer (18 marks)

Take $g = 10 \text{ m/s}^2$ and $P_{\text{atm}} = 1 \text{ atm} = 10^5 \text{ Pa}$

- | | | | | |
|--|--|-------------|---------------|-------------------|
| 17- On a day when the temperature reaches 77 °F. The temperature in degrees Celsius is | (a) 196.2 | (b) 60.6 | (c) <u>25</u> | (d) 81 |
| 18- A steel rod of 40 cm long at 40 °C is heated to 70°C. The final length is | $\alpha = 11 \times 10^{-6} \text{ }^\circ\text{C}^{-1}$ (for steel) | | | |
| | (a) <u>40.0132 cm</u> | (b) 39.9 cm | (c) 0.0132 cm | (d) None of these |

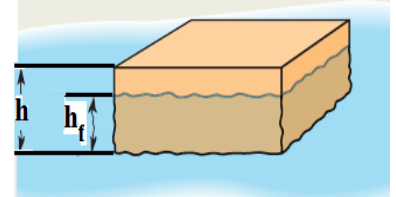
Module B4

19- A certain metal rod has a length of 10 m at 100 °C and a length of 10.04 m at 500 °C. Its length at 300 °C is
(a) 10.01 m (b) 9.99 m (c) 10.02 m (d) 9.98 m

20- If a liquid of density 1000 kg/m³. The gauge pressure at h = 50 m is
(a) 6 atm (b) 50000 Pa (c) 5 atm (d) 60000 Pa

21- A metal has a weight of 5 N is fully immersed in a fluid of density $\rho_f = 870 \text{ kg/m}^3$, then a metal weight is decreased by 2 N. The volume of the metal is
(a) 435 m³ (b) $5.75 \times 10^{-3} \text{ m}^3$ (c) $2.3 \times 10^{-4} \text{ m}^3$ (d) None of these

22- In a Fig. a block of density $\rho_o = 1000 \text{ kg/m}^3$ floats face down in a fluid of density $\rho_f = 1200 \text{ kg/m}^3$. The block has 6 cm height. The height of the block immersed is



(a) 4 cm (b) 5 cm (c) 4.8 cm (d) 6 cm

Good Luck

Aden University Faculty of Engineering Center of Preparatory Year		
Subject: Physics I Time: 1.5 hours	Test Module A3	Day: 5 / 7 /2021 Year: 2020-2021

Name:	Group:	Reg. No.
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Part I: For each statement, mark T for true and F for false (16 marks)

- | | |
|---|--------------------|
| 1- The pressure is zero as a temperature reached – 273 °C | (<u>T</u>) (F) |
| 2- If two objects are not in thermal equilibrium, they must have the different temperatures | (<u>T</u>) (F) |
| 3- Generally, liquids expand more than gases for the same temperature change | (T) (<u>F</u>) |
| 4- The zero point of Kelvin scale is above the zero point of Fahrenheit scale | (T) (<u>F</u>) |
| 5- The weight of the displaced fluid is directed downward | (<u>T</u>) (F) |
| 6- Buoyant force depends on the volume and the density of the fluid | (<u>T</u>) (F) |
| 7- The Si unit of pressure is N/m ² | (T) (<u>F</u>) |
| 8- The buoyant force changes as the depth in a fluid changes | (T) (<u>F</u>) |

Part II: Select the correct answer from the following sentences (16 marks)

- | | | | | |
|---|--|--------------------------|-----------------------------------|--------------------------|
| 9- the Si unit of a mass is | (a) m ³ | (b) N | (c) <u>kg</u> | (d) Pascal |
| 10- The ratio of the input to the output pressures in Pascal law is | (a) <u>Equal to 1</u> | (b) More than 1 | (c) Less than 1 | (d) None of these |
| 11- The energy transferred due to a temperature difference is | (a) Temperature | (b) <u>Heat</u> | (c) Pressure | (d) None of these |
| 12- A transfer of heat when no physical contact is needed is called | (a) Conduction | (b) <u>Radiation</u> | (c) Convection | (d) None of these |
| 13- A conversion from Celsius to Kelvin is | (a) T _C - 273 | (b) T _C × 273 | (c) <u>T_C + 273</u> | (d) None of these |
| 14- A shape of object determines the object is | (a) Floating | (b) Sinking | (c) Moving | (d) <u>None of these</u> |
| 15- A hydraulic lever is based on | (a) <u>Pascal principle</u> | (b) Archimedes principle | (c) Zeroth principle | (d) Calorimeter |
| 16- The mass of the displaced liquid by object is given by | (a) <u>ρ_f V_f</u> | (b) m _o g | (c) ρ _o V _o | (d) m _f g |

Part III: Solve the problems to choose the correct answer (18 marks)

Take g = 10 m/s² and P_{atm} = 1 atm = 10⁵ Pa

- | | | | | |
|---|---------------------|----------------|---------------|-------------------|
| 17- On a day when the temperature reaches 5 °F. The temperature in Kelvin scale is | (a) 48.6 | (b) <u>258</u> | (c) -15 | (d) None of these |
| 18- A steel rod of 40 cm long at 40 °C is heated to 90°C. The length should be increased by
α = 11 × 10 ⁻⁶ °C ⁻¹ (for steel) | (a) <u>0.022 cm</u> | (b) 39.9 cm | (c) 40.022 cm | (d) None of these |

Module A3

19- Metal and Aluminum are at the same lengths at 20 °C, when they are heated the same temperature, the metal increased by 0.0101 cm and the Aluminum increased by 0.022 cm. The metal is

- (a) Steel (b) Copper (c) Brass (d) None of these

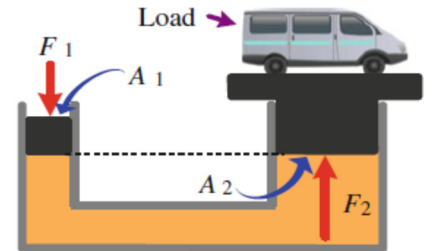
20- To suck liquid of density 900 kg/m³ up a straw to a maximum height of 4.0 m. The minimum gauge pressure must you produce is

- (a) – 36000 Pa (b) 36000 Pa (c) - 40000 Pa (d) None of these

21- A metal has a weight of 5 N is fully immersed in a fluid of density $\rho_f = 840 \text{ kg/m}^3$, then a metal weight is decreased to 2 N. The volume of the metal is

- (a) $5.952 \times 10^{-4} \text{ m}^3$ (b) 420 m^3 (c) $3.571 \times 10^{-4} \text{ m}^3$ (d) None of these

22- For the car lift shown. The pistons on the left and right have areas 25 cm² and 75 cm² respectively. The car has a total weight of 1200 N. The force must be applied on the input piston is



- (a) 40 N (b) 400 N (c) 450000 N (d) 5000 N

Good Luck

Aden University Faculty of Engineering Center of Preparatory Year		
Subject: Physics I Time: 1.5 hours	Test Module A4	Day: 5 / 7 /2021 Year: 2020-2021

Name:	Group:	Reg. No.
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Part I: For each statement, mark T for true and F for false (16 marks)

- | | |
|--|-------------|
| 1- Heat can be transferred when $\Delta T = 0$ | (T) (F) |
| 2- If two objects are not in thermal equilibrium, they must have the same temperatures | (T) (F) |
| 3- Generally, liquids expand less than gases for the same temperature change | (T) (F) |
| 4- The zero point of Kelvin scale is above the zero point of Celsius scale | (T) (F) |
| 5- The buoyant force is always directed upward | (T) (F) |
| 6- Buoyant force is independent of the shape of the object | (T) (F) |
| 7- The SI unit of pressure is not N/m^2 | (T) (F) |
| 8- The buoyant force changes as the weight of the displaced fluid changes | (T) (F) |

Part II: Select the correct answer from the following sentences (16 marks)

- | | | | | |
|---|-----------------------------|---------------------------------|------------------------------------|---------------------------|
| 9- the SI unit of a volume is | (a) <u>m^3</u> | (b) cm^3 | (c) kg | (d) Pascal |
| 10- The ratio of the input to the output pressures in Pascal law is | (a) Not equal to 1 | (b) More than 1 | (c) Less than 1 | (d) <u>None of these</u> |
| 11- The energy transferred due to a temperature difference is | (a) Temperature | (b) <u>Heat</u> | (c) Pressure | (d) None of these |
| 12- A transfer of heat when physical contact is needed is called | (a) <u>Conduction</u> | (b) Radiation | (c) Convection | (d) None of these |
| 13- A conversion from Kelvin to Celsius is | (a) $T_C - 273$ | (b) $T_C \times 273$ | (c) $T_C + 273$ | (d) <u>None of these</u> |
| 14- A shape of object determines the object is | (a) Floating | (b) Sinking | (c) Moving | (d) <u>None of these</u> |
| 15- Floating and sinking are based on | (a) Pascal principle | (b) <u>Archimedes principle</u> | (c) Zeroth principle | (d) Engineering principle |
| 16- The mass of the object displaces a fluid is given by | (a) $\rho_f V_f$ | (b) $m_o g$ | (c) <u>$\rho_o V_o$</u> | (d) $m_f g$ |

Part III: Solve the problems to choose the correct answer (18 marks)

Take $g = 10 \text{ m/s}^2$ and $P_{\text{atm}} = 1 \text{ atm} = 10^5 \text{ Pa}$

- | | | | | |
|---|----------------|-------------|----------------------|-------------------|
| 17- On a day when the temperature reaches -4°F . The temperature in Kelvin scale is | (a) <u>253</u> | (b) -20 | (c) -64.8 | (d) None of these |
| 18- A steel rod of 40 cm long at 40°C is heated to 70°C . The length should be increased by
$\alpha = 11 \times 10^{-6} \text{ }^\circ\text{C}^{-1}$ (for steel) | (a) 40.0132 cm | (b) 39.9 cm | (c) <u>0.0132 cm</u> | (d) None of these |

Module A4

19- Metal and Aluminum are at the same lengths at 20 °C, when they are heated the same temperature, the metal increased by 0.01742 cm and the Aluminum increased by 0.022 cm. The metal is

- (a) Brass** (b) Copper (c) Steel (d) None of these

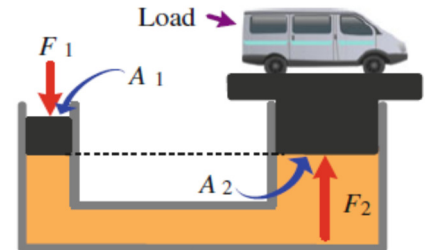
20- To suck liquid of density 900 kg/m³ up a straw to a maximum height of 0.4 m. The minimum gauge pressure must you produce is

- (a) – 3600 Pa** (b) 3600 Pa (c) - 40000 Pa (d) None of these

21- A metal has a weight of 5 N is fully immersed in a fluid of density $\rho_f = 870 \text{ kg/m}^3$, then a metal weight is decreased to 2 N. The volume of the metal is

- (a) 435 m³ (b) $5.75 \times 10^{-3} \text{ m}^3$ **(c) $3.45 \times 10^{-4} \text{ m}^3$** (d) None of these

22- For the car lift shown. The pistons on the left and right have areas 25 cm² and 750 cm² respectively. The car has a total weight of 1200 N. The force must be applied on the input piston is



- (a) 40 N** (b) 400 N (c) 500 N (d) 5000 N

Good Luck

Aden University
Faculty of Engineering
Center of Preparatory Year

Subject: Physics I
Time: 1.5 hours

Test
Module C4

Day: 6 / 7 /2021
Year: 2020-2021

Name:

Group:

Reg. No.

Part I: For each statement, mark T for true and F for false (16 marks)

- | | |
|---|--------------------|
| 1- In fluid at rest, The pressure is exerted equally in all directions | (<u>T</u>) (F) |
| 2- Celsius scale is based on properties of the water | (<u>T</u>) (F) |
| 3- The weight of the object is constant , it does not change with gravity | (T) (<u>F</u>) |
| 4- An example of radiation is the heat transferred by the waves | (<u>T</u>) (F) |
| 5- The pressure due to a rest fluid is called hydrostatic pressure | (<u>T</u>) (F) |
| 6- A pressure is not affected by a shape of the container | (<u>T</u>) (F) |
| 7- Heating by convection can occur only through gases | (T) (<u>F</u>) |
| 8- A Kelvin scale is based on a correct zero | (<u>T</u>) (F) |

Part II: Select the correct answer from the following sentences (16 marks)

- | | | | | |
|---|----------------------|-------------------------|-------------------------|--------------------------|
| 9- A temperature change in Fahrenheit as compared to a change in Celsius is | (a) Less | (b) <u>More</u> | (c) Same | (d) None of these |
| 10- In a thermometer, the thermistor should be | (a) Tall | (b) Very Large | (c) <u>Very Small</u> | (d) Short |
| 11- A change in resistance due to a temperature change occurred in | (a) Gas thermometer | (b) Mercury thermometer | (c) Alcohol thermometer | (d) <u>None of these</u> |
| 12- When the hot metal is placed in a water, some energy (heat) by water is | (a) Lost | (b) Melt | (c) <u>Gained</u> | (d) None of these |
| 13- If the ratio of the object to the fluid densities is less than one, the object is | (a) <u>Moving up</u> | (b) Sinking | (c) At rest | (d) Floating |
| 14- A measurement unit of the weight is | (a) m | (b) <u>N</u> | (c) kg | (d) m ² |
| 15- A full or partially immersed object in a fluid is buoyed up | (a) Calorimetry | (b) Pascal | (c) Zeroth law | (d) <u>Archimedes'</u> |
| 16- In Pascal application, the pressures should be | (a) Closed | (b) Rest | (c) <u>Same</u> | (d) Open |

Part III: Solve the problems to choose the correct answer (18 marks)

Take $g = 10 \text{ m/s}^2$ and $P_{\text{atm}} = 1 \text{ atm} = 10^5 \text{ Pa}$

- | | | | | |
|---|------------------|-----------|---------------|-------------------|
| 17- The change of temperature is 30 K. The equivalent change in Fahrenheit scale is | (a) 30 | (b) 16.7 | (c) <u>54</u> | (d) None of these |
| 18- A 50 g sample of copper is at 30 °C. If 1200 J of energy is added to it by heat, The final temperature of the copper is ($c = 387 \text{ J/kg.}^\circ\text{C}$) | (a) <u>92 °C</u> | (b) 62 °C | (c) 32 °C | (d) None of these |

Module C4

-
- 19- Two metals with different length are L_1 and L_2 have $\alpha_1 = 10 \times 10^{-6} \text{ } ^\circ\text{C}^{-1}$ and $\alpha_2 = 14 \times 10^{-6} \text{ } ^\circ\text{C}^{-1}$.
If $L_1 - L_2 = 12 \text{ cm}$ at any temperatures. Then L_1 is equal to
(a) 48 cm (b) 36 cm (c) 27 cm **(d) 42 cm**
-
- 20- A 5 kg object is released from rest while fully immersed in a liquid. The liquid displaced has a mass of 2 kg.
The acceleration of the object is
(a) 6 m/s² (b) 4 m/s² (c) 4.6 m/s² (d) None of these
-
- 21- A 50 N object floats in liquid $\rho_f = 10^3 \text{ kg/m}^3$. If the object is immersed by 4 cm and object's height above a surface of liquid is 1 cm. The volume of the object should be
(a) $8.3 \times 10^{-3} \text{ m}^3$ **(b) $6.25 \times 10^{-3} \text{ m}^3$** (c) $7.5 \times 10^{-3} \text{ m}^3$ (d) None of these
-
- 22- A piece of metal of $m_o = 0.5 \text{ kg}$ and a density $\rho_o = 8.7 \times 10^3 \text{ kg/m}^3$. The metal is immersed in fluid of density $\rho_f = 10^3 \text{ kg/m}^3$. The weight of the metal in fluid is
(a) 4.36 N (b) 5 N **(c) 4.4 N** (d) None of these
-

Good Luck